

Bihar Engineering University, Patna**B.Tech. 1st Semester Examination, 2023****Course: B.Tech.****Time: 03 Hours****Code: 100105****Subject: Workshop Manufacturing Practices****Full Marks: 70****Instructions:-**

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only):**[2 × 7 = 14]**

- (a) The cutting edge of a chisel should be
 - (i) Hardened
 - (ii) Tempered
 - (iii) Hardened and Tempered
 - (iv) Case Hardened
- (b) The purpose of a riser is to
 - (i) Deliver molten metal into mould cavity
 - (ii) Act as a reservoir for molten metal
 - (iii) Feed the molten metal to the casting in order to compensate for the shrinkage
 - (iv) Deliver the molten metal from pouring basin to gate
- (c) The limitation of electro-chemical machining (ECM) process is
 - (i) the use of corrosive media as electrolyte makes it difficult to handle
 - (ii) poor surface finish
 - (iii) poor accuracy of work piece dimensions because of large tool wear
 - (iv) there will be thermal damage to the work piece
- (d) In forehand welding, the weld is made
 - (i) from left to right
 - (ii) from right to left
 - (iii) first from left to right then from right to left
 - (iv) either from left to right or from right to left
- (e) Which type of hammer is used for extracting the nails out of wood?
 - (i) Ball peen hammer
 - (ii) Claw peen hammer
 - (iii) Cross peen hammer
 - (iv) Straight peen hammer
- (f) Which of the following processes of moulding is widely used for the manufacturing of bottle caps and automotive dashboards?
 - (i) Compression moulding
 - (ii) Transfer moulding
 - (iii) Injection moulding
 - (iv) Jet moulding
- (g) Why are the plasticisers added to polymers?
 - (i) To hold other constituents of plastic together.
 - (ii) To reduce the cost and enhance the strength and hardness of plastics.
 - (iii) To improve flexibility and to reduce the temperature and pressure required for moulding of plastics.
 - (iv) None of the above.
- (h) Which material is commonly used in consumer-grade 3D printers?
 - (i) Metal
 - (ii) Glass
 - (iii) Plastic
 - (iv) Rubber
- (i) In forging process, metals are shaped by ____
 - (i) Impact
 - (ii) Cohesion
 - (iii) Tension
 - (iv) None of the mentioned
- (j) The liquid temperature of the filler metal used in brazing is ____
 - (i) 150°C
 - (ii) 427°C
 - (iii) 723°C
 - (iv) 1000°C

- Q.2** (a) What do you understand by plant layout and shop layout? List out the equipment required for a foundry shop and draw the shop layout. [7]
- (b) Write a note on need and development of additive manufacturing system [7]
- Q.3** (a) Enumerate the safety precautions to be taken in a workshop. [7]
- (b) Distinguish between thermoplastic and thermosetting plastic. [7]
- Q.4** (a) Explain the meaning of metal forming. Give the classification of metal forming processes. [6]
- (b) What is the difference between hot forming and cold forming? Discuss any three metal forming processes with neat sketches. [8]
- Q.5** (a) What is a pattern? Name various allowances associated with pattern and why are they provided? [7]
- (b) Discuss the different types of defects in casting process. [7]
- Q.6** (a) Make a sketch of centre lathe and label its main parts. [8]
- (b) Explain any three operations performed on a lathe machine with neat sketches. [6]
- Q.7** (a) Sketch the quick return motion mechanism used in shaper and explain why it is provided in it. [7]
- (b) List out the different types of files. Write the use of any three types of files with neat sketch. [7]
- Q.8** (a) Describe the principle of oxyacetylene gas welding with neat sketch. [7]
- (b) Distinguish between welding, brazing and soldering. Which process produces the strongest joints and why? [7]
- Q.9** Write short notes on the following:- [13]
- (a) CNC machine
 - (b) Power tools
 - (c) Seasoning of wood
 - (d) Glass cutting

